

Arina Veprikova

Canada (open to relocation) | Available May 2026

ava75@sfu.ca | +1 (778) 586-4810 | [linkedin.com/in/arina-veprikova-a97526366](https://www.linkedin.com/in/arina-veprikova-a97526366) | areenve.github.io

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Java, MATLAB

Analytics & ML: scikit-learn, tidymodels, caret, statsmodels, logistic regression, classification, threshold tuning, cross-validation, out-of-sample evaluation, feature engineering

Medical & Statistical Analysis: pandas, NumPy, scipy, seaborn, matplotlib, imbalanced-learn (class imbalance handling), ROC/AUC analysis, sensitivity-specificity optimization, calibration (Platt scaling / isotonic), epidemiological-style screening evaluation

Data & Systems: SQLite (schema design, queries), BigQuery (analytics + forecast workflows), Git/GitHub, Quarto/R Markdown

BI & Visualization: Power BI, Tableau, plotly, ggplot2, TradingView (Pine Script overlays)

EDUCATION

Simon Fraser University (SFU)

Burnaby, BC

BSc, Data Science — Expected 2028

3rd year

Selected coursework: Database Systems (CMPT 354), Data Structures (CMPT 225), Linear Algebra, Linear Optimization, Business Statistics, Introduction to Data science (STAT240).

EXPERIENCE

ICBC (Insurance Corporation of British Columbia)

Vancouver, BC

Change Management Co-op

Apr 2026 – Aug 2026

- Maintain initiative-level inputs for the enterprise **Change Saturation Index (CSI)** dashboard — validating weighted CSI scores and organizational capacity data used in executive reporting.
- Consolidate cross-divisional people-impact data from Change Specialists into reusable senior-leadership reporting templates: data tables, visual summaries, and slide content aligned to ICBC brand standards.
- Document stakeholder & employee impacts for digital workplace initiatives, including SharePoint intranet transition, Hub redesign, Copilot, and Teams Premium pilots, feeding change impact assessments.
- Support the Head Office Relocation project with research, documentation, and stakeholder analysis, including DEI Action Plan considerations; draft PSWG communications under senior Change Management Specialist direction.

LLC “AuditDelo”

Irkutsk, Russia

Junior Accountant / Data Analyst

Feb 2022 – Dec 2024

- Automated recurring audit workpapers in **Excel**, including templates, validations, and controls, cutting preparation time per organization by **~40%** and improving traceability.
- Processed & QA-checked **~10 organizations/week** of audit documentation; reconciled exceptions and partnered with auditors and finance teams to submit final packs on time.
- Built reusable Excel templates with drop-down logic, embedded validation checks, and standardized documentation structures for recurring client files.

PROJECTS

NHANES Undiagnosed Diabetes Screening Model + Shiny App (*R, tidymodels, Shiny, survey*)

[GitHub](#)

- Built a population-level **diabetes screening system** using NHANES 2017–2020 data (**9,232 individuals**), incorporating survey weights to estimate real-world prevalence (**~10.3%**).
- Defined a clinically grounded target for **undiagnosed diabetes** ($\text{HbA1c} \geq 6.5\%$, no diagnosis/medication), identifying **284 cases** with **~2.4% prevalence**.
- Engineered clinically interpretable features (BMI categories, waist risk, age interactions) and applied **probability calibration** and **threshold optimization**, reducing decision threshold from 0.50 to **0.03**, achieving **85.9% sensitivity** in a rare-event setting.
- Demonstrated screening tradeoff: **45% sensitivity / 80% specificity** at a practical operating point, aligning model behavior with public health screening objectives.
- Developed an interactive **Shiny application** that converts model outputs into real-time risk predictions, enabling user-facing screening decisions.

Market Inefficiencies Pipeline (*Python, SQLite, TradingView*)

[GitHub](#)

- Built a crypto cross-venue dislocation pipeline to detect short-lived price gaps after fees, spreads, and slippage filters.
- Stored detected events in SQLite and summarized event count, duration, and peak spread distributions using Python visualizations.
- Created TradingView Pine Script overlays to visually compare detected Python signals against market price behaviour.